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(a)	<i>Newton's law of gravitation</i> states that every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
(b)(i)	Gravitational force provides the centripetal force.
	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $\text{Kinetic energy} = \frac{1}{2}mv^2$ $= \frac{GMm}{2r}$
(b)(ii)	Total energy = Kinetic energy + Gravitational potential energy $= \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$
	$= -\frac{GMm}{2r}$
(c)(i)	Period of rotation of satellite, $T = 24.6 \times 3$ $= 73.8 \text{ hours}$
	Gravitational force of Mars on satellite provides the centripetal force. $\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2$ $r^3 = \frac{(6.67 \times 10^{-11})(6.39 \times 10^{23})(73.8 \times 60 \times 60)^2}{4\pi^2}$
	$r = 4.24 \times 10^7 \text{ m (3 s.f.)}$
(c)(ii)	Work done $= -\frac{GMm}{2(2r)} - \left(-\frac{GMm}{2r}\right)$ $= \frac{1}{2}\left(\frac{GMm}{2r}\right)$ $= \frac{(6.67 \times 10^{-11})(6.39 \times 10^{23})(470)}{4(4.24 \times 10^7)}$
	$= 118 \text{ MJ (3 s.f.)}$
(d)	Total energy will be reduced. This means that the radius of circular orbit will gradually decrease.
	Decreasing circular orbit means that the kinetic energy increases and thus the speed of the satellite would increase.

Anglo-Chinese Junior College
H2 (9749) Physics

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